

Parth Thakre

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EDUCATION

University of Illinois Urbana-Champaign

Jan 2026 – May 2027

Master's in Computer Science

GPA: 4.00

Purdue University - West Lafayette

Aug 2022 – Dec 2025

Bachelor of Science in Computer Science and Data Science

GPA: 3.69

- Relevant Coursework: Artificial Intelligence, Machine Learning, Algorithms, Object Oriented Programming, Data Structures, Probability, Data Engineering, Computer Architecture, Statistical Theory, Operating Systems

EXPERIENCE

The Data Mine – Purdue University

West Lafayette, IN

Undergraduate Data Science Researcher

Aug 2025 – Dec 2025

- Worked in an agile, sprint-based environment with Microsoft, contributing to iterative development cycles
- Improved development efficiency by 2x by prototyping and testing Minecraft add-ons using vibe coding practices

Hewlett Packard Enterprise

Spring, TX

Data Science Intern

May 2025 – Aug 2025

- Built a full-stack team management system (Node.js API + React UI) integrated with LDAP for 5,000+ users, delivering secure RBAC and streamlined CSV-based bulk operations.
- Implemented access control interface with Sequelize ORM and Redux thunks managing 20,000+ product mappings
- Designed REST API with server-side caching and pagination for large-scale enterprise datasets

Purdue Vertically Integrated Projects (VIP)

West Lafayette, IN

Undergraduate Research Assistant

Aug 2024 – Aug 2025

- Developed coefficient of variation-based algorithm reducing AP-10K data by 82% while improving Average Precision by 5%, accelerating RTMPose training from 2.5-3 hours to 25-30 minutes
- Implemented keypoint labeling pipeline with Label Studio to enhance pose detection accuracy

Purdue University

West Lafayette, IN

Undergraduate Researcher

Aug 2024 – Aug 2025

- Analyzed 35+ years of IMPROVE aerosol data from Midwest sites using plots for time series, regressions, Mann-Whitney U and Mann-Kendall tests to identify spikes in aerosol compositions due to wildfires
- Developed 2000+ line Python pipeline to quantify OC:EC ratio trends utilizing statistical testing

PROJECTS

AI Risk Assessment | *React, Next.js, FastAPI, Python, SQLAlchemy, LLM*

- Designed and implemented a 3-stage LLM security pipeline with FastAPI asyncio across 10+ endpoints
- Utilized async system to retrieve prompts from the database, send to LLM, and automatically grade using LLMs
- Built a full-stack platform using Next.js + Tailwind CSS frontend with FastAPI + SQLAlchemy backend APIs

Bull-It | *React, FastAPI, LLM, Python, XGBoost*

- Engineered ML stock predictor achieving 84% accuracy using XGBoost, LSTM, RSI, MACD, Bollinger Bands
- Trained and validated models on 20+ years of market data using robust train, validation, and test splits
- Built full-stack application with FastAPI backend, React frontend, and AWS S3 for secure storage and retrieval

PUBLICATIONS

Improving Animal Pose Estimation through Species Similarity Measures and Rigorous Label Definition

Mar 2026

Co-Author; IEEE Winter Conference on Applications of Computer Vision (WACV)

TECHNICAL SKILLS

Languages: Python, Java, C++, R, SQL, JavaScript (Node.js, React), SAS, HTML/CSS

Machine Learning & AI: TensorFlow, PyTorch, scikit-learn, XGBoost, Hugging Face, VADER, RTMPose

Data Analysis & Visualization: pandas, NumPy, Matplotlib, Seaborn, ggplot2, Tableau, PyQt6

Tools & Platforms: FastAPI, REST APIs, AWS S3, SSMS, Git/GitHub, Docker, Linux, SAP, LabelStudio